

What a domain prior can and cannot replace

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TGDC case study, research track

Can hand-written rules stand in for a client’s operational records when generating enterprise data? On 311 tickets from four cities, blind rules clear a structural null in San Francisco (p 0.0015) and not in Austin (p 0.110), and a conditional fitted on the real data beats them in both. Two cheaper things do most of the work: category volume alone scores 0.672 against my own rules’ 0.676, and a city’s published service target scores 79% of the achievable margin with no reasoning and no records at all. What survives is a warning rather than a substitute. If a taxonomy plus a published target carries most of the signal, an agent benchmark built on generated tickets is testing something much weaker than it looks.

Companies selling generated enterprise data build it from a description of a business rather than anyone’s real records: an org chart, a list of the work, and hand-written rules for how that work behaves. A client who cannot share their data can still be served, which is the appeal. What it costs is the open question. Distributional fidelity is easy to measure and already known to be a poor guide to downstream usefulness, so the practical question is which properties of real records the rules can supply and which a client has to hand over.

This study uses public 311 service request feeds from four cities. Each ticket is a unit of work and the question asked of it is how long it takes to close. A service catalogue lists job types and the departments that own them with no timings, so it stands in for the description of a business, and the recorded durations stand in for the client records a generator does without.

Hypothesis

My hypothesis is that rules reasoned from the nature of the work, written without sight of any outcome, predict how long a job takes at an AUC between 0.60 and 0.72, and that anything below 0.55 would mean the reasoning adds nothing.

Related work

That fidelity and utility can come apart is established [1, 2, 3, 4], and whether the standard metrics measure the right thing is itself argued over [5]. I take that as given rather than as something to show, and the fidelity ladder below is here only because it makes the point on the same data as everything else.

The nearer literature is language models as priors over tabular problems, and this is a small instance of it. Knauer et al. induce a decision tree from the target and the column names alone, with no rows and no labels [6], which is the move my model-written rules make, and TabLLM shows column names carry zero-shot signal before any label [7]. AutoElicit scores an elicited prior against real data by marginal likelihood and chooses between priors by Bayes factor [8], which does not depend on a test split and is what I would adopt next time.

All four are classification where my target is duration, and the first three need feature names to be descriptive, which Knauer et al. state as a precondition. A 311 taxonomy is unusually kind on that point. An enterprise schema of coded columns would not be.

Method

The records are written by the people who do the work. Each feed gives the work type, the owning department, how the request came in, and when it opened and closed. I take the full weeks of 4 May and 8 June 2026 (Table 1).

Tickets still open when I pulled the data are kept and counted as slow, since all are months past any threshold. Chicago closes most of its feed within a second of creation, and I drop those rows as records rather than work.

The task is to predict, at the moment a ticket opens, whether it will take longer than the median to close. Recovered skill, reported alongside AUC, is $(\text{AUC} - 0.5) / (\text{ceiling} - 0.5)$. The null I test against is that the rules carry no information about which work runs slow, so any score they reach is available from carving up the same taxonomy at random.

The ceiling: a prior reads two columns, the service category and the owning department, so I fit the conditional rate from those same two to see what they support. It is a taxonomy ceiling, not a general limit on skill, and it separates a weak rule from weak features. I settled on it after the first Austin arm had been scored against a different estimator, so the AUCs here are pre-registered and the denominator is not. Every percentage sits beside its raw AUC.

I report two kinds of interval. Resampling tickets treats 17,007 San Francisco rows as independent when the rules were written per category, so I also resample categories [9]. That second unit carries two warnings. A score is constant within a category only where the category has one owning department, which holds throughout Austin but fails for 22 of San Francisco’s 37 categories, 99.6% of its rows. And clustered resampling is asymptotic in the number of clusters and assumes rough balance, where one San Francisco category holds 34% of the rows and the effective count is 5.3 against 37 categories, Austin 22.3 against 117. Almost nothing could clear an interval that wide, so I read the category intervals as descriptive rather than as tests, and where I say something is not established I mean this design cannot establish it [10].

I pulled each catalogue with the query limited to category names and row counts, so no duration, rate or outcome reached me. I wrote the rules from that alone, committed them with a prediction, then evaluated. I amended one San Francisco file 27 seconds after committing it, to match on word boundaries rather than substrings, before that city’s data existed on disk.

The biggest risk is that the result describes one person’s reasoning, so Austin and San Francisco each have two sets

city	train	test	median resolution	categories
New York	72,206	78,296	5.4h	164
Chicago	17,291	38,671	118.0h	92
Austin	5,968	6,029	23.8h	117
San Francisco	17,723	17,007	15.1h	37

Table 1: the four feeds, two complete weeks from each.

of rules under the same blind condition. I wrote one, a language model wrote the other from the same catalogue. New York is the contrast, written while I could see the real rates, and Chicago is where I carried those rules.

Results

What a blind prior recovers

The answer is not yes or no (Table 2). The best blind rules recover 72% of the taxonomy ceiling in San Francisco, the weakest nothing in Austin. Had I tested one city I would have drawn a confident conclusion either way, and the city would have decided which.

The two intervals disagree. By ticket three of the four arms beat chance, but by category only the two San Francisco arms do, and Austin’s model-written set spans [0.442, 0.770], which I cannot tell from guessing. The 72% spans roughly 7% to 93% on that unit, so read every claim below at that width.

What the rules add over structure alone

A rule set can score well because it is reasoning, or because almost any sensible cut of a lopsided taxonomy would. Two controls tell those apart.

Shuffling the verdicts across categories keeps the taxonomy, the row counts and the set of scores, and breaks only which category got which score, so it draws from the null directly and the share of shuffles reaching the observed AUC is a permutation p-value (Table 3). The null assigns one verdict per category, so the arm is scored that way too, which is why the table carries a second AUC for it. The rules reach 0.741 in San Francisco with 2 of 2,000 shuffles matching, p of 0.0015, and 0.617 in Austin with 220 matching, p of 0.110. I reject the null in San Francisco and cannot in Austin, so the reasoning does real work in one city and is not distinguishable from a lucky cut of the taxonomy in the other.

Frequency is the harder baseline, because the pull gave me row counts and those are themselves real operational data. Scored rarer-is-slower it reaches 0.672 against 0.676 for my own San Francisco rules, so mine add almost nothing beyond volume. The model’s clear it by 0.105. Picking that direction is one bit the catalogue did not give me, so the baseline is generous if anything.

The author effect is not established

The model beats me by 0.116 in Austin and 0.101 in San Francisco, the same direction and a similar size twice, which looks like something. By ticket both miss zero, at [0.104, 0.131] and [0.095, 0.107], but by category neither does, at [-0.036, 0.288] and [-0.021, 0.174]. Two cities are not enough, and an earlier draft of mine reported these differences with no interval at all.

What I predicted, before I looked

arm	predicted	observed	in band
Austin, mine	0.60 to 0.72	0.494	no
Austin, the model	0.52 to 0.62	0.611	yes
SF, mine	0.55 to 0.68	0.676	yes
SF, the model	none	0.777	-

I set the first band before the project had any result, and it was wrong. I set the next two after seeing Austin, so I knew roughly what to expect, which is a weaker achievement and I count it as one. The fourth arm produces my headline 72% and carries no prediction at all. The only prediction made in real ignorance is the one that failed.

One department explains most of the gap

Austin Resource Recovery (Table 4) is 34% of the city’s volume and 70% of its work runs slow, yet both sets of rules call it fast, 0.37 for the model and 0.05 for mine. Take it out and the model reaches 70% against San Francisco’s 72% and I reach 38% against 46%, so the gap closes for both of us. A missed collection sits open until the next scheduled route, a week away. The work is routine. The ticket is not, and nothing in the phrase ARR - Compost tells you so.

Run on San Francisco the same operation cuts the other way, which is the more important half of the table. Its two largest categories are 57% of test volume, both fast and both called fast. Drop them and the ceiling barely moves, 0.886 to 0.851, while the model falls from 72% to 27% and I fall from 46% to 20%. A two-line rule calling those two fast and the rest slow scores 0.699 by itself, so the 72% rests on a few high-volume decisions, not 37 categories of reasoning.

ARR is a department prefix rather than a category called waste, so removing it also takes out fast work it owns, such as ARR - Dead Animal Collection at 0.0% slow. And Graffiti Public is 8.2% of San Francisco volume and 85.1% slow but priced at 0.37 by the model, so the failure ARR shows in Austin is present there too and no subset here removes it.

What the prior does not beat

Given the same two columns, a conditional fitted on the real data beats the blind rules by 0.108 in San Francisco and 0.272 in Austin, so real records win on this task in both cities. Against a conditional using the category column alone the rules look level, 0.777 to 0.771, but that comparator is missing the department column the rules read, which is worth 0.115 in San Francisco, and the 0.0067 margin has a paired interval of [-0.201, +0.108].

The finding

Stated as an ordering rather than a level, because the level moves with which categories sit in the subset and the interval on it is wide. Blind rules beat chance in San Francisco and not in Austin, the model’s beat mine in

city	rules	AUC	by ticket	by category	ceiling	recovered
Austin	mine	0.494	[0.481, 0.508]	[0.331, 0.649]	0.883	-1%
Austin	the model	0.611	[0.598, 0.626]	[0.442, 0.770]	0.883	29%
San Francisco	mine	0.676	[0.670, 0.684]	[0.515, 0.815]	0.886	46%
San Francisco	the model	0.777	[0.770, 0.784]	[0.529, 0.859]	0.886	72%

Table 2: what blind rules recover, by city and by author. Ticket intervals are 1,000 resamples, category intervals 2,000.

city	arm	AUC	[95% CI]	recovered	p vs H0
Austin	blind rules, per ticket	0.611		29%	
Austin	blind rules, per category	0.617			0.110
Austin	category frequency alone	0.516	[0.351, 0.726]	4%	
Austin	same rules, verdicts shuffled	0.501	[0.322, 0.674]	0%	
SF	blind rules, per ticket	0.777		72%	
SF	blind rules, per category	0.741			0.0015
SF	category frequency alone	0.672	[0.441, 0.785]	45%	
SF	same rules, verdicts shuffled	0.499	[0.312, 0.690]	0%	

Table 3: blind rules against two baselines built from structure alone, with category intervals and the permutation test of the null.

both cities, and removing one administratively clocked department closes most of the gap between the cities for both authors.

My explanation for that ordering, that a blind prior holds where duration follows from the job and fails where the clock is an administrative cycle, is a hypothesis and not a result. I derived the partition from which categories my rules got wrong, in one city, so it cannot fail against the data that produced it. The test that would settle it is in the closing section.

What the real rates are worth

Rules written with a city’s rates in hand reach 81% there, above the 72% a blind author got in San Francisco but not by much.

The last row I did not expect. New York publishes the target it holds itself to, in days, per complaint type. Ranking tickets by that alone reaches 0.830, or 79%, matching rules written while looking at the real outcomes and needing no reasoning, no language model and no client records. If a client publishes what they intend to take, that document is worth about as much as the exercise this paper measures, and it is the first thing to ask for.

It bears on contamination too, since a language model may be recalling published material rather than reasoning, and this row shows such a document carries most of the signal. I could not find an equivalent dataset for Austin or San Francisco, which narrows that worry without settling it.

Carried to Chicago at Chicago’s own median they score 0.496, which an earlier draft called actively misleading. That does not hold up (Table 5): the medians are 5.4h and 118.0h, so the carried row asks a question nobody wrote the rules for. At New York’s own threshold they reach 0.581, or 37%. What is left is mechanical, since they fire on 17.1% of Chicago rows, being New York acronyms.

Distribution fidelity misses all of this

My first recorded hypothesis, fixed in `HYPOTHESIS.md` before I ran anything, was that generated records matching real data to low-order fidelity would still hold combinations the real process could never produce, and that accuracy would fall as those rose. I tested it by fitting generators on real New York records, sampling, training

on the sample and testing on real records [11]. It failed, for a structural reason. My pairwise generator draws department conditional on category, so it cannot emit a pair it never saw and its 0.0% impossible rate is an identity, and the learner is additive over one-hot columns with no interaction terms, so a wrong combination could not have hurt it either. Both were settled by the design before any data was touched. The work moved to the narrower question this paper answers, which is why my stated hypothesis and my pre-registration are not the same sentence.

The ladder still shows the one thing it is here for. Marginals alone lose almost everything, 0.511 against a 0.918 ceiling, while adding pairwise dependence recovers all of it at 0.917. A generator clearing the column shapes and column pair trends a quality report scores [12] is indistinguishable from real records here, so a distribution-level check cannot see the effect this paper measures. My generators are conditional probability tables with no privacy noise, so they are not PrivBayes or MST, but the order of structure they keep is the order those methods fit [13, 14].

Limits

Four cities, one domain, one task, and only two cities blind. New York is a contamination control and Chicago its transfer target, so neither replicates anything, and what does replicate is an author effect whose category intervals cross zero. A 311 feed has three usable columns, no documents that must agree with each other and no prices, so it is a thin stand-in for enterprise data.

The second author is a language model, the biggest hole in my design. These feeds are widely mirrored, so blind means the session showed it no durations, not that the weights hold none. Models have memorised popular tabular datasets and score better on ones they have seen [15], the exact failure this arm is open to, and it produces every headline number. I did not record the prompt or the model version, so I cannot rerun it.

I wrote the San Francisco rules knowing what Austin had shown, so they are blind to that city’s rates but I am not blind to the lesson.

The Austin model arm has an ordering problem I should state. The commit that evaluated my own rules names

subset	rules	n	prior	ceiling	recovered
Austin, all	mine	6,029	0.494	0.883	−1%
Austin, all	the model	6,029	0.611	0.883	29%
Austin, minus ARR	mine	4,008	0.644	0.881	38%
Austin, minus ARR	the model	4,008	0.766	0.881	70%
SF, all	mine	17,007	0.676	0.886	46%
SF, all	the model	17,007	0.777	0.886	72%
SF, minus top 2	mine	7,255	0.571	0.851	20%
SF, minus top 2	the model	7,255	0.595	0.851	27%

Table 4: dropping the highest-volume categories in each city, for both sets of rules. The same operation run both ways.

rules	tested on	threshold	AUC	ceiling	recovered
New York	New York	5.4h	0.840	0.918	81%
NY rules carried	Chicago	118.0h	0.496	0.717	−2%
NY rules carried	Chicago	5.4h	0.581	0.717	37%
published target	New York	5.4h	0.830	0.918	79%

Table 5: rules written with the rates in hand, those rules carried to another city, and the city’s own published service target.

three categories with their measured slow rates: compost at 96%, traffic signal maintenance at 8%, vehicle abatement at 94%. My rules call all three the wrong way. The model wrote its Austin rules twelve hours later and got two of the three right, using my own phrasing as keyword strings. Compost it still called fast, which argues against the rates being handed over wholesale, and both corrections are argued for in the file on grounds that do not need them. I did not record the prompt, so I cannot settle it. The blind condition I describe covers the catalogue pull, not what I had already written down by the time that author was asked. The ARR result is a subset I chose after seeing which family my rules got wrong, so it fits the numbers and nothing has tested it. New York carries known artefacts too, including a department that closes most tickets at exactly midnight.

What it would take to answer the general question

Make the rules generate. My arms score rule sets against a conditional fitted on real data, which measures how much of a conditional a person can guess, whereas the question asks what generated data costs. Sampling rows from the taxonomy and volume counts, labelling them with the prior, then training on those and testing on real records would answer it directly. Then predict the failures before looking: my ARR explanation says any category whose clock is an administrative cycle will break a blind prior, so name those in a fifth city from the taxonomy alone, record the list, then measure.

This bears on evaluating agents. If a taxonomy and a published target reach most of the achievable skill, an agent scored on tickets generated from that taxonomy is being scored on the part of the problem that was easiest to reconstruct. The categories where a blind prior fails are the ones a generated benchmark will get wrong, and they are invisible in the taxonomy it was built from. A high score there is evidence about the generator, not the agent.

A prior can only be evaluated as a prior once per dataset, before anyone has seen the outcomes, and the prediction has to come first. Of the predictions I made

here, the only one made in real ignorance is the one that turned out wrong.

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